Functional recovery after ischemic stroke—A matter of age

Data from the Austrian Stroke Unit Registry

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ABSTRACT

Objective: To analyze the association between patient age and good functional outcome after ischemic stroke with special focus on young patients who were numerically underrepresented in previous evaluations.

Methods: Of 43,163 ischemic stroke patients prospectively enrolled in the Austrian Stroke Unit Registry, 6,084 (14.1%) were ≤55 years old. Functional outcome was available in a representative subsample of 14,256 patients free of prestroke disability, 2,223 of whom were 55 years or younger. Herein we analyzed the effects of age on good functional outcome 3 months after stroke (modified Rankin Scale score ≤2).

Results: Good outcome was achieved in 88.2% (unadjusted probability) of young stroke patients (≤55 years). In multivariable analysis, age emerged as a significant predictor of outcome independent of stroke severity, etiology, performance of thrombolysis, sex, risk factors, and stroke complications. When the age stratum 56–65 years was used as a reference, odds ratios (95% confidence interval [95%CI]) of good outcome were 3.4 (1.9–6.4), 2.2 (1.6–3.2), and 1.5 (1.2–1.9) for patients aged 18–35, 36–45, and 46–55 years and 0.70 (0.60–0.81), 0.32 (0.28–0.37), and 0.18 (0.14–0.22) for those aged 66–75, 76–85, and >85 years ($p < 0.001$). In absolute terms, the regression-adjusted probability of good outcome was highest in the age group 18–35 years and gradually declined by 3.1%–4.2% per decade until age 75 with a steep drop thereafter. Findings applied equally to sexes and patients with and without IV thrombolysis or diabetes.

Conclusions: Age emerged as a highly significant inverse predictor of good functional outcome after ischemic stroke independent of stroke severity, characteristics, and complications with the age-outcome association exhibiting a nonlinear scale and extending to young stroke patients.
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GLOSSARY

mRS = modified Rankin Scale; **NIHSS** = NIH Stroke Scale.

Stroke was long considered a paradigmatic disease of elderly people, yet population-based evaluations demonstrated that more than 20% of ischemic strokes affect working-age people. Approximately 8% and 3% occur before the ages of 55 and 45 years, respectively.[1,2] In specialized stroke units, up to 12% of patients are younger than 45 years.[3] It is well documented that mortality after ischemic stroke is low in the young.[4,5] However, information on functional outcome is sparse in this age group and mainly derived from single center studies enrolling several hundred patients or less.[3,6-9] Paucity of this particular information is in obvious contrast to the high economic and clinical relevance of stroke in young individuals. Conversely, the steadily rising life expectancy results in an increase in stroke-related disability in elderly and old-old patients. Therefore, the present report analyzes the association between patient age and

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Supplemental Data

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good functional outcome after ischemic stroke in the large multicenter registry of Austrian stroke units with a special focus on the young and evaluates age-dependent changes in the characteristics of stroke patients.

METHODS Since 2003, a growing network of Austrian stroke units covering the whole country (Austrian Stroke Unit Network) prospectively collected data on standard characteristics and acute management of all stroke patients admitted to 33 of 34 Austrian stroke units. Data collection and ratings were performed by experienced stroke neurologists using standardized variable definitions and scores. Evaluations were done at the time of admission to and discharge from the stroke unit and supplemented by a 3-month follow-up call. To ensure high data quality, immediate data entry was obligatory, and the Web-based database featured online plausibility checks and help. In biannual meetings of stroke neurologists from the participating centers, details about scoring procedures and variable assessment are thoroughly discussed. Details on the registry have been reported previously.[10,11] The modified Rankin Scale (mRS) score was assessed on admission and discharge from the stroke unit, as well as at the 3-month follow-up. The NIH Stroke Scale (NIHSS) score was evaluated on admission to the stroke unit. Good functional outcome was defined as mRS score ≤2 on the 3-month follow-up visit.

Hypertension, diabetes mellitus, and hyperlipidemia were diagnosed according to the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure,[12] the World Health Organization criteria,[13] and the criteria of the National Cholesterol Education Program.[14] Atrial fibrillation, if not already detected before stroke, was diagnosed on the admission EKG or during EKG monitoring on the stroke unit. All risk conditions were coded at the time of discharge from the stroke unit. Whenever an accurate classification could not be determined at this time, status was coded as unknown.

Regular drinking was defined as consumption of more than 250 mL of wine or 500 mL of beer per day on at least 5 days a week.

Standards for the diagnosis of clinical complications were suggested according to previous evaluations in the acute stroke setting.[15,16] In brief, pneumonia was suspected upon respiratory crackles, fever, or purulent sputum and confirmed by chest X-ray. Typical clinical symptoms combined with a positive urine dipstick examination for nitrite, pyuria, or positive urine cultures indicated urinary tract infection. Clinically relevant intracranial bleeding episodes were rated according to National Institute of Neurological Disorders and Stroke criteria (any CT or MRI documented bleeding with clinical deterioration of at least 1 point on the NIHSS or leading to death within 7 days).[17] Cardiac decompensation was considered as a condition of symptomatic heart failure requiring acute medical intervention and characterized by dyspnea, venous congestion, and pulmonary edema. Clinical worsening was defined as an increase by ≥2 points on the NIHSS during stroke unit care.[10]

Information on the sex distribution in the Austrian population was supplied by Statistik Austria (www.statistik.at). The female/male ratio was calculated by comparing the actual number of women and men with acute ischemic stroke in a specific age stratum with the number of women and men living in Austria in the same age stratum: age-specific ratio = (women with stroke/women living in Austria)/(men with stroke/men living in Austria). Stroke patients not living in Austria were not included in this calculation.

Statistical analysis. As a statistical environment, R (version 2.11.1)[18] was used. The association between age and good functional outcome was analyzed by means of logistic regression analysis. The multivariable model was built by a stepwise selection procedure (optimizing the Akaike information criteria), allowing for the following variables: age categories (18–35, 36–45, 46–55, 56–65, 66–75, 76–85, and 86–103 years), sex, stroke severity measured by the NIHSS, prestroke disability (mRS), hypertension, diabetes mellitus, hyperlipidemia, current smoking status, regular drinking, atrial fibrillation, previous stroke, myocardial infarction, peripheral artery obstructive disease, acute cardiac decompensation on the stroke unit, pneumonia, urinary tract infection, clinically relevant intracranial bleeding, IV thrombolysis, and stroke etiology (as defined by the Trial of Org 10172 in Acute Stroke Treatment [TOAST] classification). First-order interactions between those variables were calculated to estimate effect modification between the individual risk predictors. Regression-adjusted probabilities of good functional outcome were calculated using the marginal prediction method.[19] Information on the age and sex structure of the Austrian population (population pyramid) was derived from Statistik Austria (www.statistik.at).

Standard protocol approvals, registrations, and patient consents. The registry is part of a governmental quality assessment program for stroke care in Austria financed by the Federal Ministry of Health. It is based on the federal law promoting quality in health (Gesundheitsqualitätsgesetz). Anonymized data are centrally administered by the Gesundheit Österreich GmbH, and scientific analyses are approved and supervised by an academic review board.

RESULTS Between March 1, 2003, and January 18, 2011, 43,163 adult patients with acute ischemic stroke were recorded in the Austrian Stroke Unit Registry. Patients' characteristics are summarized in the table. In a total of 2,273 and 6,084 patients, stroke manifested at an age of 18–45 and 18–55 years, respectively (figure e-1 on the *Neurology*® Web site at www.neurology.org). There was a female preponderance among stroke patients older than age 75 (figure 1). However, after considering the age and sex structure of the Austrian population (population pyramid), women were less likely to have a stroke over the whole age range except for individuals of 35 years and younger (figure 1). Frequencies of vascular risk factors in different age groups are depicted in figure e-2. The frequency of atrial fibrillation among stroke patients increased substantially with age from 1.3% (18–35 years) to 46.3% (>85 years).

Information on functional outcome at 3 months was available in a representative subsample of 16,453 patients, and 14,256 of them were functionally independent before stroke (mRS score ≤2). Characteristics of this subsample were similar to those of the entire registry. When patients with prestroke disability were excluded, those with and without 3-month follow-up showed significant (Bonferroni adjusted

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Table Characteristics of stroke patients in the Austrian Stroke Unit Registry (2003-2011)

Characteristic	Ischemic stroke (n = 43,163)	Ischemic stroke with 3-month follow-up (n = 14,256)
Age, median (IQR)	73 (63–81)	71 (61–79)
Male, n (%)	22,679 (52.5)	7,858 (55.1)
NIHSS, median (IQR)		
Admission	4 (2–9)	3 (1–8)[a]
Discharge	2 (0–5)	1 (0–4)[a]
mRS score, median (IQR)		
Prestroke	0 (0–1)	0 (0–0)[a]
Admission	3 (1–4)	3 (1–4)[a]
Discharge	2 (1–4)	2 (0–4)[a]
Follow-up	1 (0–3)	1 (0–3)
IV thrombolysis, n (%)	4,892 (12.1)	1,890 (13.3)
Risk factors, n (%)		
Hypertension	31,640 (77.9)	10,864 (76.1)[a]
Diabetes mellitus	10,349 (25.5)	3,330 (23.4)
Hyerlipidemia	20,619 (50.7)	7,465 (52.4)
Atrial fibrillation	10,593 (26.1)	3,389 (23.8)
Smoking	7,225 (17.8)	2,847 (20.0)[a]
Regular drinking	3,370 (8.3)	1,188 (8.3)[a]
Previous stroke	9,818 (24.2)	2,799 (19.6)[a]
Previous myocardial infarction	3,773 (9.3)	1,236 (8.7)
Complications on stroke unit, n (%)		
Pneumonia	2,323 (5.4)	697 (4.9)[a]
Urinary tract infection	1,820 (4.2)	585 (4.1)[a]
Cardiac decompensation	940 (2.2)	321 (2.3)[a]
Clinically relevant intracerebral bleeding	565 (1.3)	235 (1.6)[a]
Clinical worsening	2,669 (6.9)	810 (6.0)

Abbreviations: mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale.
[a] Bonferroni adjusted $p < 0.05$ for difference between subgroups with and without 3-month follow-up (restricted to those without prestroke disability [mRS score ≤2]).

$p < 0.05$), albeit very small, differences for NIHSS score upon admission and discharge, mRS score (prestroke, admission, and discharge), frequency of hypertension, smoking, regular drinking, and previous stroke as well as for frequency of complications (pneumonia, urinary tract infection, cardiac decompensation, and clinically relevant intracerebral bleeding) (table). Distributions of mRS scores of 0–6 in each age group are summarized figure 2. Good outcome (mRS score ≤2) was common among young stroke patients (unadjusted probabilities, 92.1% and 88.2% in the age ranges 18–45 and 18–55 years). In multivariable analysis, age emerged as a significant inverse predictor of good outcome independent of stroke severity and various other characteristics (figure 3). The scale of association was nonlinear (figure 4). In brief, up to age 75, the regression-adjusted probability of good functional outcome decreased in absolute values by 3.1%–4.2% for each 10-year increase in chronologic age, whereas after age 75 the probability of good functional outcome steeply declined by approximately 10% per 10-year increment in age. On top of age and sex, stroke severity (NIHSS score on admission), preexisting disabilities (prestroke mRS score), performance of IV thrombolysis, current smoking, diabetes mellitus, and stroke complications (pneumonia, urinary tract infections, cardiac decompensation, and clinical relevant intracerebral bleeding) all significantly predicted 3-month functional outcome and were considered in the multivariable analysis. Additional consideration of stroke etiology, previous stroke, myocardial infarction or other cardiac diseases, atrial fibrillation, peripheral artery occlusive disease, regular alcohol consumption, hypertension, or hyperlipidemia had no effect on the results obtained. When the analysis was restricted to individuals aged 18–55 years, age still emerged as a highly significant predictor of outcome ($p < 0.001$). The association between age and functional outcome applied equally to men and women, to patients with and without diabetes, and to those treated and not treated with IV thrombolysis (p for effect modification >0.05 each) (figure 4). Furthermore, this association was independent of stroke etiology (data not shown).

Mortality on stroke units was as low as 0.5% in patients 18–55 years of age and mainly stroke-related. After discharge from the stroke unit follow-up at 3 months, an additional 37 (1.5%) patients died due to reasons as follows: intracerebral edema, 12 (0.5%); recurrent stroke, 2 (0.1%); myocardial infarction, 2 (0.1%); cardiac diseases, 3 (0.1%); tumors 6, (0.2%); and other known, 3 (0.1%) or unknown diseases, 9 (0.4%).

DISCUSSION The current evaluation includes the largest sample of young stroke patients published so far and allows accurate assessment of age effects on functional outcome from youth to old-old. As a key finding, age inversely predicted good functional outcome 3 months after stroke over the whole age range (figure 3). This association was independent of stroke severity and other characteristics such as risk factors, acute stroke therapy, complications, and stroke etiology and extended to men and women and patients with diabetes and IV thrombolysis. Of note, the association was nonlinear with a gradual decline in the adjusted absolute probability of good outcome by 3.1%–4.2% per decade up to age 75 and a steep drop thereafter (figure 4). In young individuals, a favorable functional outcome was achieved in approximately 80%–90% of patients, which agrees with previous evaluations reporting a good functional outcome in

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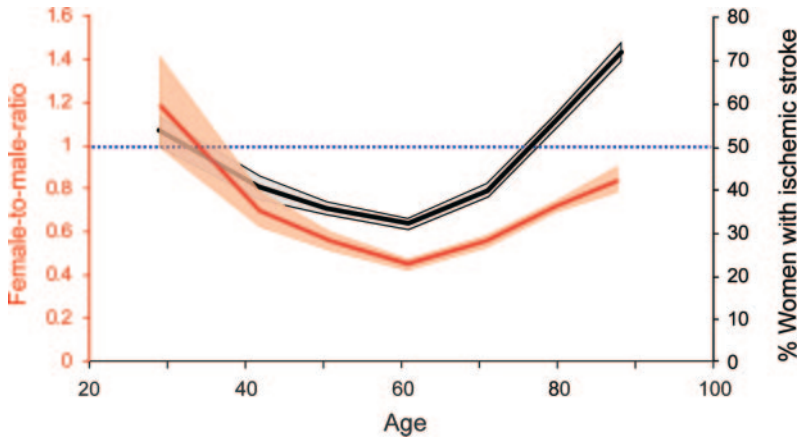
Figure 1 Gender distribution of ischemic stroke patients in Austrian stroke units (n = 43,163)



Black lines represent the percentage (95% confidence interval [CI]) of woman among all stroke patients on Austrian stroke units, and the red line depicts the female/male ratio (95% CI) corrected for the age and sex structure (population pyramid) of the Austrian population.

between 53.0% and 92.7% of young stroke patients, depending on the length of the observation period, the definition of young, and the cutoff chosen for classification of outcome.[3,6-9,20-24] These data, however, have to be interpreted in light of several reports suggesting persistently reduced quality of life and high frequency of fatigue after ischemic stroke in individuals in this age range. Moreover, only about 60% of young stroke patients return to working life.[6,8,24-26] This discrepancy was also reported for patients with subarachnoid hemorrhage and probably reflects persistent deficits not captured by the mRS such as mood disturbances and cognitive dysfunction.[27]

Better recovery in young stroke patients is not unexpected. Potential explanations are intact vascular compensation mechanisms and a more pronounced neuronal plasticity. Our findings in young stroke patients fit very well to results from experimental stroke models in rats indicating that young animals have a better functional outcome than older littermates for an identical lesion size of stroke.[28,29] These experiments clearly document an enhanced regenerative capacity in the brain of young rats as well as an age-dependent deregulation of neuronal and glial repair and impaired angioneogenesis.[29-31] Because in Austria both hospital stay and rehabilitation are fully covered by the health insurance, it is not likely that quality and access to treatment and rehabilitation substantially confound our findings. Data from the registry actually indicate high and rather constant proportions of stroke patients with inpatient rehabilitation up to an age of 75–85. Moreover, comorbidities may contribute to the worse outcome in elderly individuals (figure 4). This issue was partly addressed by adjusting the analyses for baseline mRS score, morbid conditions such as diabetes and severe heart failure, and stroke complications. Finally, the steep decline in the probability of good outcome in elderly individuals may partly reflect the rapidly increasing frequency of poststroke cognitive deficits and dementia at higher ages.[32]

Our large multicenter evaluation is confirmatory to previous studies showing that men face a higher risk of stroke at any age except for individuals aged

Figure 2 Three-month functional outcome after ischemic stroke assessed by means of the modified Rankin Scale (mRS) score according to age groups

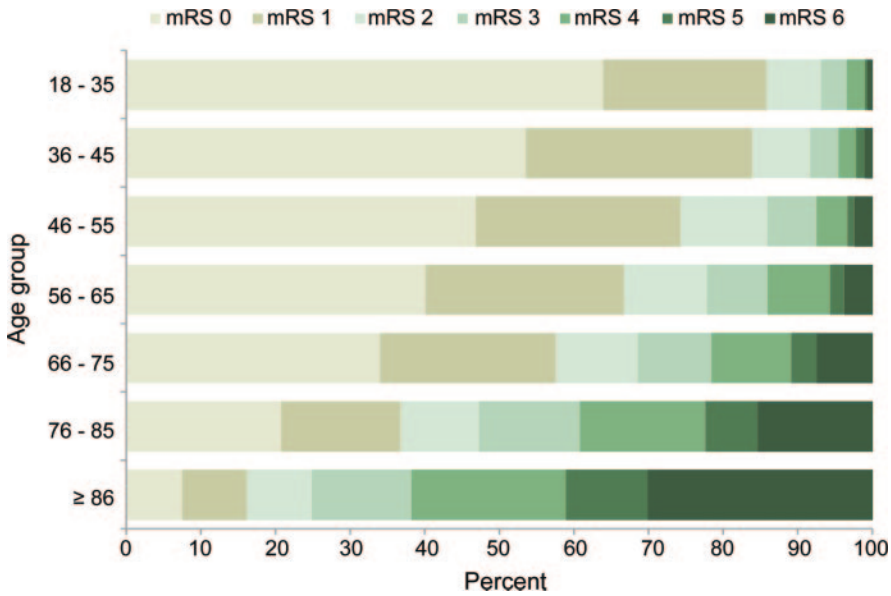


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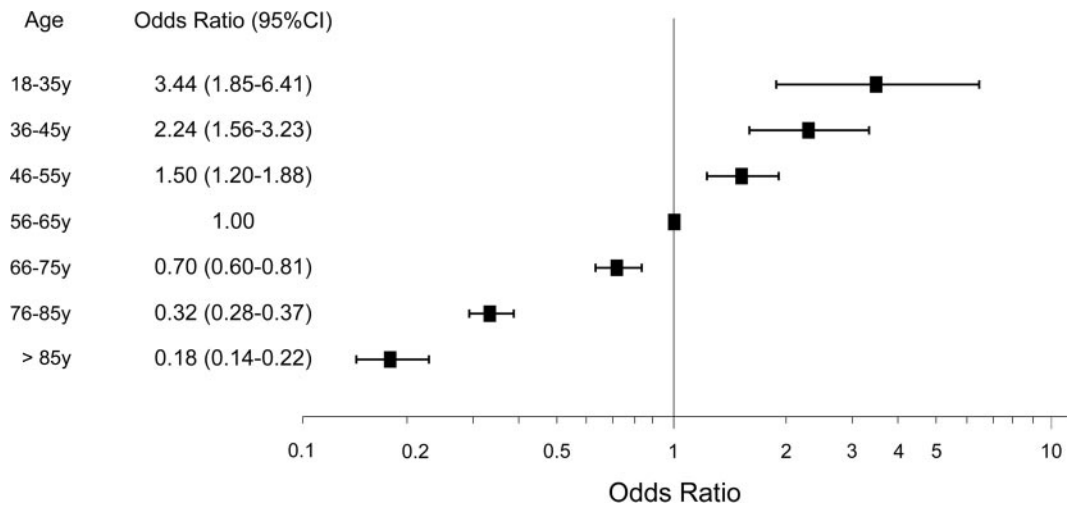
Figure 3 Odds ratios (95% confidence interval [CI]) of a good functional outcome (modified Rankin scale [mRS] score ≤2) 3 months after stroke in different age groups

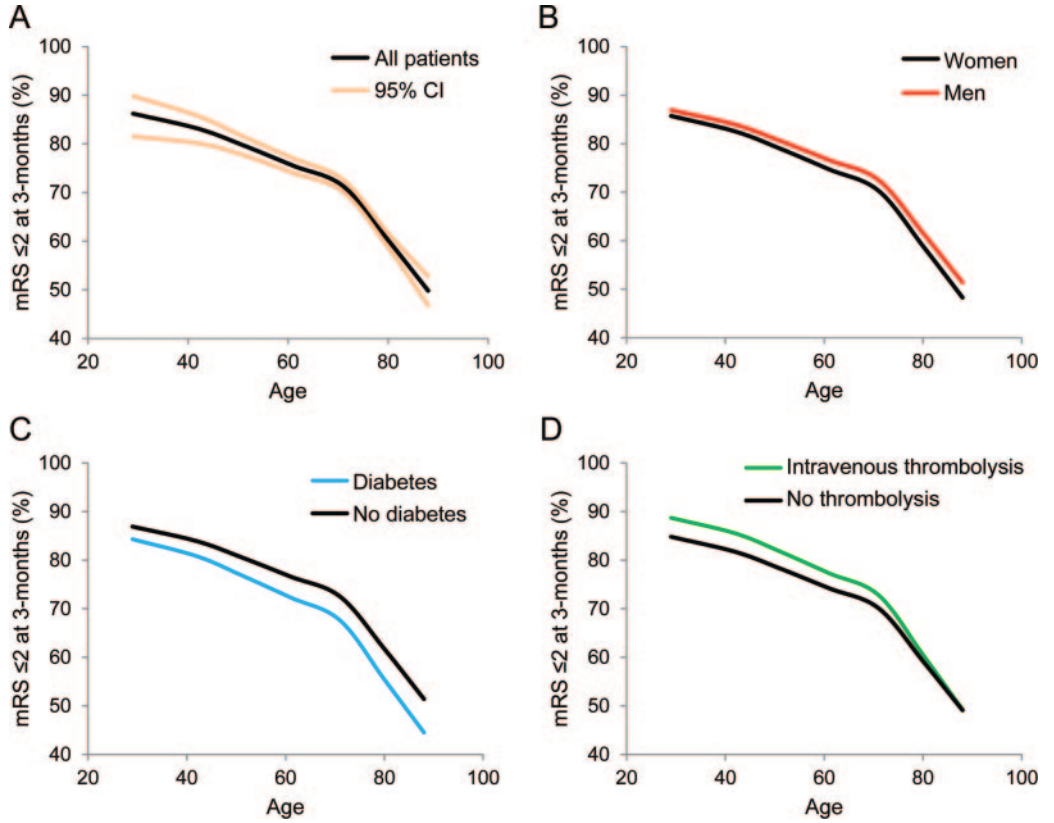


Results were derived from multivariable logistic regression analysis adjusted for sex, stroke severity (NIH Stoke Scale score on admission), preexisting disabilities (prestroke mRS score), IV thrombolysis, current smoking, diabetes mellitus, and stroke complications (pneumonia, urinary tract infections, cardiac decompensation, and clinical relevant intracerebral bleeding).

35 and younger (figure 1) and the proportion of strokes attributable to atrial fibrillation substantially increases with age (figure e-2).

The main strengths of our study are its considerable size (n = 43,163) and representativity for a whole country (33 of 34 Austrian stroke units are

Figure 4 Regression-adjusted probability of good functional outcome (modified Rankin scale [mRS] score ≤2) 3 months after stroke in various age groups



For adjustment see figure 3. (A) Whole population. (B) Men and women. (C) Patients with and without diabetes. (D) Patients with or without IV thrombolysis. CI = confidence interval.

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participating). Data were collected prospectively, underwent a rigorous quality control, and included a structured follow-up and assessment of 3-month functional outcome. However, one has to keep in mind that our registry, although spanning the whole country, is stroke unit– and not population-based. Currently, approximately two-thirds of all ischemic stroke patients are treated in the participating stroke units and entered in the registry, whereas patients who died before hospital admission and those treated on general hospital wards including patients with severe comorbidities such as cancer were not. The information available in our database is limited for practical reasons, as there is no way to reassess additional data because of the protection of anonymity. Furthermore, the 90-day follow-up was available in a subgroup only.

Age is a key determinant of functional outcome after ischemic stroke independent of stroke severity and other characteristics, and this finding extends to young stroke patients. It is tempting to speculate about age-dependent differences in neuronal plasticity, repair capacity, vascular architecture, and comorbidities as mechanisms underlying this association. The discrepancy between good functional outcome in our study and striking consequences on quality of life and social circumstances (divorce rates and early retirement) in previous reports deserves further research and calls for increased awareness of the multimodal burdens of stroke in young individuals. Both young stroke patients with commonly mild deficits and a long life span ahead and elderly patients, clearly outnumbering the young patients, with more severe disability and reduced life expectancy pose severe socioeconomic burdens to the society.

AUTHOR CONTRIBUTIONS

Dr. Knoflach designed and conceptualized the study, interpreted the data, and drafted the manuscript. Dr. Matosevic interpreted the data and revised the manuscript. Dr. Rücker interpreted the data and revised the manuscript. Dr. Furtner interpreted the data and revised the manuscript. Dr. Mair interpreted the data and revised the manuscript. Dr. Wille interpreted the data and revised the manuscript. Dr. Zangerle interpreted the data and revised the manuscript. Dr. Werner interpreted the data and revised the manuscript. Dr. Ferrari interpreted the data and revised the manuscript. Dr. Schmidauer interpreted the data and revised the manuscript. L. Seyfang analyzed the data. Prof. Kiechl designed and conceptualized the study, interpreted the data, and drafted the manuscript. Prof. Willeit interpreted the data and revised the manuscript.

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REFERENCES

1. Bamford J, Sandercock P, Dennis M, Burn J, Warlow C. A prospective study of acute cerebrovascular disease in the community: the Oxfordshire Community Stroke Project: 1981–86. 2. Incidence, case fatality rates and overall outcome at one year of cerebral infarction, primary intracerebral and subarachnoid haemorrhage. J Neurol Neurosurg Psychiatry 1990;53:16–22.
2. Kolominsky-Rabas PL, Sarti C, Heuschmann PU, et al. A prospective community-based study of stroke in Germany: the Erlangen Stroke Project (ESPro): incidence and case fatality at 1, 3, and 12 months. Stroke 1998;29:2501–2506.
3. Nedeltchev K, der Maur TA, Georgiadis D, et al. Ischaemic stroke in young adults: predictors of outcome and recurrence. J Neurol Neurosurg Psychiatry 2005;76:191–195.
4. Putaala J, Curtze S, Hiltunen S, et al. Causes of death and predictors of 5-year mortality in young adults after first-ever ischemic stroke: the Helsinki Young Stroke Registry. Stroke 2009;40:2698–2703.
5. Marini C, Totaro R, Carolei A. Long-term prognosis of cerebral ischemia in young adults: National Research Council Study Group on Stroke in the Young. Stroke 1999;30:2320–2325.
6. Leys D, Bandu L, Henon H, et al. Clinical outcome in 287 consecutive young adults (15 to 45 years) with ischemic stroke. Neurology 2002;59:26–33.
7. Naess H, Nyland HI, Thomassen L, Aarseth J, Myhr KM. Long-term outcome of cerebral infarction in young adults. Acta Neurol Scand 2004;110:107–112.
8. Varona JF, Bermejo F, Guerra JM, Molina JA. Long-term prognosis of ischemic stroke in young adults: study of 272 cases. J Neurol 2004;251:1507–1514.
9. Arnold M, Halpern M, Meier N, et al. Age-dependent differences in demographics, risk factors, co-morbidity, etiology, management, and clinical outcome of acute ischemic stroke. J Neurol 2008;255:1503–1507.
10. Ferrari J, Knoflach M, Kiechl S, et al. Early clinical worsening in patients with TIA or minor stroke: the Austrian Stroke Unit Registry. Neurology 2010;74:136–141.
11. Ferrari J, Knoflach M, Kiechl S, et al. Stroke thrombolysis: having more time translates into delayed therapy: data from the Austrian Stroke Unit Registry. Stroke 2010;41:2001–2004.
12. Chobanian AV, Bakris GL, Black HR, et al. The seventh report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure: the JNC 7 report. JAMA 2003;289:2560–2572.
13. Definition, Diagnosis and Classification of Diabetes Mellitus and Its Complications: Report of a WHO Consultation, Part 1: Diagnosis and Classification of Diabetes Mellitus. WHO/NCD/NCS/99.2. Geneva, World Health Organization; 1999.
14. Third report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III) final report. Circulation 2002;106:3143–3421.
15. Davenport RJ, Dennis MS, Wellwood I, Warlow CP. Complications after acute stroke. Stroke 1996;27:415–420.
16. Langhorne P, Stott DJ, Robertson L, et al. Medical complications after stroke: a multicenter study. Stroke 2000;31:1223–1229.
17. Tissue plasminogen activator for acute ischemic stroke: the National Institute of Neurological Disorders and Stroke rt-PA Stroke Study Group. N Engl J Med 1995;333:1581–1587.
18. R Development Core Team. R: A Language and Environment for Statistical Computing. Vienna, Austria: R Foun-

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dation for Statistical Computing; 2010. Available at: www.R-project.org. Accessed December 6, 2011.
19. Wilcosky TC, Chambless LE. A comparison of direct adjustment and regression adjustment of epidemiologic measures. J Chronic Dis 1985;38:849–856.
20. Marini C, Totaro R, De Santis F, et al. Stroke in young adults in the community-based L'Aquila registry: incidence and prognosis. Stroke 2001;32:52–56.
21. Patella R, Spalloni A, Ferrari M, et al. Cerebral ischemia in young patients (under 45 years of age): clinical and neuroradiological follow-up. Neurol Sci 2011;32:427–432.
22. Naess H, Nyland HI, Thomassen L, et al. Incidence and short-term outcome of cerebral infarction in young adults in western Norway. Stroke 2002;33:2105–2108.
23. Spengos K, Vemmos K. Risk factors, etiology, and outcome of first-ever ischemic stroke in young adults aged 15 to 45: the Athens Young Stroke Registry. Eur J Neurol 2010;17:1358–1364.
24. Kappelle LJ, Adams HP Jr, Heffner ML, et al. Prognosis of young adults with ischemic stroke: a long-term follow-up study assessing recurrent vascular events and functional outcome in the Iowa Registry of Stroke in Young Adults. Stroke 1994;25:1360–1365.
25. Naess H, Nyland HI, Thomassen L, Aarseth J, Myhr KM. Fatigue at long-term follow-up in young adults with cerebral infarction. Cerebrovasc Dis 2005;20:245–250.
26. Naess H, Waje-Andreassen U, Thomassen L, Nyland H, Myhr KM. Health-related quality of life among young adults with ischemic stroke on long-term follow-up. Stroke 2006;37:1232–1236.
27. Rinkel GJ, Algra A. Long-term outcomes of patients with aneurysmal subarachnoid haemorrhage. Lancet Neurol 2011;10:349–356.
28. Zhang L, Zhang RL, Wang Y, et al. Functional recovery in aged and young rats after embolic stroke: treatment with a phosphodiesterase type 5 inhibitor. Stroke 2005; 36:847–852.
29. Popa-Wagner A, Carmichael ST, Kokaia Z, Kessler C, Walker LC. The response of the aged brain to stroke: too much, too soon? Curr Neurovasc Res 2007;4:216–227.
30. Badan I, Buchhold B, Hamm A, et al. Accelerated glial reactivity to stroke in aged rats correlates with reduced functional recovery. J Cereb Blood Flow Metab 2003;23: 845–854.
31. Petcu EB, Sfredel V, Platt D, et al. Cellular and molecular events underlying the dysregulated response of the aged brain to stroke: a mini-review. Gerontology 2008;54:6–17.
32. Pendlebury ST, Rothwell PM. Prevalence, incidence, and factors associated with pre-stroke and post-stroke dementia: a systematic review and meta-analysis. Lancet Neurol 2009;8:1006–1018.

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